FNUG: Factor-Neutral Update Gating for Anchored Option Hedging

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Abstract

Daily option chains contain rich cross-sectional information, but classical delta-vega hedges summarize the market through only a few local sensitivities. Neural policies can exploit the broader quote set, but unconstrained policies may overtrade and depart from classical delta-vega controls. We propose Factor-Neutral Update Gating (FNUG), which learns controlled hedge-set updates around a least-squares delta-vega hedge (LSDeltaVega) from irregular sets of observed option-chain quotes. The learned quote-set state enters only through a bounded adjustment, projected toward the delta-vega null space and scaled by a trust gate. The objective penalizes both tail loss and anchor-relative underperformance, keeping the learned update accountable to the classical hedge.

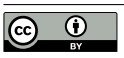
We evaluate FNUG on daily S&P 500 Index (SPX) option data from 2018–2025, using observed marks, spreads, and transaction costs across three liability families, cost levels, and random seeds. In the primary at-the-money (ATM) call setting, FNUG reduces CVaR_{95} relative to LSDeltaVega by 16.9%, a gain that remains significant under a paired moving-block bootstrap. The same bootstrap test supports improvement over the plain anchored-residual QiaoResidual-LS baseline. Across all 27 liability-cost-seed settings, FNUG has lower CVaR_{95} than LSDeltaVega. Against the strongest fixed implied-volatility (IV) grid comparator, IVGrid-HedgeSet, FNUG has lower CVaR_{95} in 22/27 matched evaluations, with an average CVaR_{95} difference of -0.0064 across matched settings. In the primary setting, FNUG also has lower trading cost and turnover than IVGrid-HedgeSet. Overall, the results support factor-neutral gated updating as a way to reduce hedging tail risk without relaxing classical delta-vega control.

CCS Concepts

• **Computing methodologies** → **Sequential decision making**; *Neural networks*; • **Applied computing** → *Economics*.

Keywords

deep hedging, option hedging, transaction costs, CVaR, tail risk, factor-neutral update gating, anchored option hedging, option-chain quotes



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ICAIF '26, Milan, AA, Italy
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ACM ISBN 979-8-4007-3087-0/2026/11
https://doi.org/10.1145/3858063.3858697

ACM Reference Format:
Tenghan Zhong, Yuxuan Ouyang, and Keyuan Wu. 2026. FNUG: Factor-Neutral Update Gating for Anchored Option Hedging. In *7th ACM International Conference on AI in Finance (ICAIF '26), November 14–17, 2026, Milan, AA, Italy*. ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3858063.3858697

1 Introduction

Option hedging is a sequential control problem under noisy market state information, limited liquidity, and transaction costs [18, 26]. Classical delta and delta-vega hedges are stable and interpretable, but they summarize the option market through only a few local sensitivities. Deep hedging methods can condition on richer market states and optimize nonlinear risk measures directly, but unconstrained neural hedges may overtrade or depart from the risk discipline of a classical delta-vega hedge. This tension is particularly relevant in historical option-chain data, where observed bid-ask spreads make excessive trading costly and the available contracts vary across dates.

Historical option-chain data also pose a coupled representation problem: the market state is an irregular set of quotes, while the hedge instruments form a variable action set. A common approach regularizes the cross section by interpolating an implied-volatility (IV) surface onto a fixed grid or compressing it into a small set of surface features [9, 10]. These representations make the state tractable, but they do not by themselves control how a learned policy deviates from a classical hedge. Conversely, trading restrictions and Conditional Value-at-Risk (CVaR) objectives control trading activity and tail risk [11, 14], but do not by themselves determine how information from an irregular quote set should enter a multi-instrument hedge. Together, these advances leave open whether a learned policy can use irregular option-chain information to improve a classical hedge without sacrificing trading discipline or tail-risk control.

We address this gap with FNUG, short for Factor-Neutral Update Gating. Its central design separates classical risk control from learned policy improvement. The deterministic least-squares delta-vega hedge (LSDeltaVega) targets first-order risk matching over the instruments available on each date, while the learned branch uses the encoded quote cross section to propose a bounded update. Factor-neutral projection suppresses the update's delta-vega exposure, and a trust gate controls how much is executed. Since the anchor remains active at every rebalance, the policy reverts exactly to LSDeltaVega when the update is suppressed. The anchor-relative objective further penalizes updates that worsen tail outcomes on the same historical episodes. In this way, FNUG casts deep hedging as controlled improvement of a classical hedge rather than unrestricted policy learning.

Our contributions are as follows.

- We formulate option hedging from quote sets to hedge sets: an irregular daily option chain defines the market state, and the hedge instruments selected at episode start define the action space. This lets the policy condition on observed quotes without first imposing a fixed IV grid.
- We develop FNUG, an architecture for factor-neutral gated updates that separates classical Greek hedging from learned hedge-set adjustment. The deterministic LSDeltaVega anchor controls delta and vega, while the neural branch uses option-chain information to propose bounded updates that approximately preserve these exposures. Training penalizes both tail loss and underperformance relative to the anchor on the same episodes, making improvement over the classical hedge an explicit learning target.
- We provide an exact-contract, quote-marked SPX evaluation across liabilities, transaction costs, and random seeds. FNUG lowers CVaR_{95} relative to LSDeltaVega in all 27 settings and has lower CVaR_{95} than IVGrid-HedgeSet in 22 of 27 matched settings. In the primary ATM-call setting, paired moving-block bootstrap supports gains over LSDeltaVega and QiaoResidual-LS.

2 Related Work

Deep hedging and data-driven hedge construction. Deep hedging formulates hedging under market frictions as direct optimization of a neural trading policy under a chosen risk measure [2]. Empirical studies learn hedges directly from historical option data [20, 24], where missing contract quotes and valuation choices can affect the losses used for evaluation [29]. Multi-option extensions use options themselves as hedge instruments [12]; other variants change the policy architecture or add interpretability [8, 25]. A complementary data-driven approach combines learned market scenarios with a separate optimization layer: Cont and Vuletić use a conditional generative model to compute hedge ratios, select instruments, and incorporate transaction costs [5]. Classical models have also been used to initialize or supervise learned hedges from market data [15, 21]. A close residual-learning precedent is Qiao and Wan, who learn a correction around the Black–Scholes delta rather than relearning the full hedge [22]. Compared with this single-underlying, one-step residual formulation, FNUG uses a multi-instrument LSDeltaVega delta-vega anchor and evaluates factor-neutral hedge-set updates by their episode-level excess risk relative to that anchor.

Option-market representations. A common way to make option cross sections tractable is to convert irregular quotes into structured implied-volatility surfaces or low-dimensional state variables. Classical surface parameterizations such as SVI and SSVI provide structured IV representations [10]. Neural-SDE models learn arbitrage-aware low-dimensional dynamics of the option book and derive hedges from the resulting state representation [4]. More directly related to our fixed-grid comparator, recent deep hedging methods use IV-surface states with multiple hedge instruments and transaction costs [9]. Operator Deep Smoothing also starts from irregular quote configurations but maps them to smooth IV surfaces [27]. These approaches impose a structured market representation before the hedge decision, using fitted factors, grids, or surface states. Deep Sets and Set Transformers instead provide permutation-invariant architectures for variable-size observations [17, 28]. FNUG uses this machinery for the hedge decision itself: the policy state is learned directly from the observed quote set, while surface-shape penalties regularize the representation without replacing the chain by a fitted surface or fixed grid.

Trading and tail-risk controls. With proportional transaction costs, classical control models produce no-trade regions around frictionless hedge positions [6, 26]. The No-Transaction Band Network brings this prior into neural hedging architectures [14]. CVaR is a coherent risk measure for adverse loss tails [1, 23], and Godin studies dynamic CVaR hedging directly under transaction costs [11]. Recent distributional reinforcement-learning methods also target gamma-vega hedging and extreme loss quantiles [3, 19]. FNUG uses CVaR in a different role: the objective penalizes tail loss and same-episode underperformance relative to an explicit LSDelta Vega anchor, while the update channel is separately constrained to keep delta-vega exposure small.

3 Historical Hedging Setup

Quote-set state. On each trading day t, the option market is observed as an irregular quote set $Q_t = \{x_{t,i}\}_{i=1}^{N_t}$, where N_t is the number of available quotes and $x_{t,i}$ is the feature vector of quote i. Each feature vector contains log-moneyness $k_{t,i} = \log(K_{t,i}/F_{t,i})$, time-to-maturity $\tau_{t,i}$, option type, total implied variance $w_{t,i} = \tau_{t,i}\sigma_{t,i}^2$, bid and ask quotes, relative spread, and liquidity features. Here $K_{t,i}$ is the strike, $F_{t,i}$ is the same-day forward to expiration, and $\sigma_{t,i}$ is the Black–Scholes implied volatility. A Set Transformer encoder with parameters θ (64-dimensional quote embeddings, two induced-attention blocks, 32 inducing points, four heads, and multihead-attention pooling) maps the set to $z_t = f_\theta(Q_t) \in \mathbb{R}^{d_z}$, where d_z is the latent dimension. Encoder–decoder training on 2018–2021 quote sets combines quote reconstruction, masked quote recovery, bid-ask coverage, and calendar/butterfly penalties, followed by fine-tuning for short-horizon volatility and stress prediction. Forecast targets crossing split boundaries are discarded. The encoder is selected by 2022 validation forecast loss and frozen before test-state extraction; policy normalizers use training episodes only, and decoder outputs are not used in the hedging backtest.

Episode and action set. Episodes have horizon H and are locally indexed by $t = 0, \ldots, H$, with observed liability midpoint marks L_t. At episode start, a rule selects up to M hedge instruments $\mathcal{H}$ from the option chain, including the underlying and liquid options or straddles. A straddle coordinate represents a paired call-plus-put position; its midpoint, half-spread, delta, and vega are the sums of the two legs. At active rebalances $t = 0, \ldots, H-1$, $P_t \in \mathbb{R}^M$ collects option-instrument midpoint marks and the underlying close, and the policy chooses positions $q_t \in \mathbb{R}^M$; P_H contains terminal marks for revaluation and liquidation. The underlying leg is a stylized liquid index exposure marked to the SPX close, with proportional cost but no quoted half-spread. If fewer than M instruments are selected, an availability mask $v_t \in \{0,1\}^M$ zeros padded positions and transaction costs.

Quote-marked P&L and evaluation metrics. Set $q_{-1} = 0$ before the episode and set $q_H = 0$ for terminal liquidation. For $t = 0, \ldots, H$,

let $d_t = q_t - q_{t-1}$ be the trade vector. Let $h_{t,j}$ be the observed half-spread of hedge instrument j at time t, with $h_{t,j} = 0$ for the underlying, and let κ be the proportional transaction cost rate. The trading cost TC_t is

$$\mathrm{TC}_t = \sum_{j=1}^{M} |d_{t,j}| \left(\kappa |P_{t,j}| + h_{t,j}\right). \tag{1}$$

For $t = H$, this formula charges the cost of closing the terminal hedge position. For a short position of one unit in the liability, the terminal episode P&L is

$$\Pi(q) = \sum_{t=0}^{H-1} q_t^\top (P_{t+1} - P_t) - \sum_{t=0}^{H-1} (L_{t+1} - L_t) - \sum_{t=0}^{H} \mathrm{TC}_t. \tag{2}$$

Let $\bar{L}_0 = \max\{L_0, 1\}$. The normalized loss is $\ell(q) = -\Pi(q)/\bar{L}_0$, and normalized transaction-cost terms use the same scale. Episode turnover is $\sum_{t=0}^{H} \|q_t - q_{t-1}\|_1$, using $q_H = 0$ for terminal liquidation. We track mean normalized loss, CVaR_{95}, CVaR_{99}, worst-regime CVaR_{95}, transaction cost, and turnover. Throughout, CVaR_{95} and CVaR_{99} denote empirical upper-tail CVaR of normalized loss at the 95% and 99% confidence levels, respectively. For $\alpha \in (0, 1)$ and a finite sample X of normalized losses, let $q_\alpha = \mathrm{quantile}_\alpha(X)$; $\mathrm{CVaR}_\alpha(X)$ averages the observations $x \in X$ satisfying $x \geq q_\alpha$. The corresponding Value-at-Risk (VaR) quantile is denoted by VaR_{95}. For episode n, let ℓ_n denote its normalized loss and let $c_n \in \{1, \ldots, G\}$ denote its regime label. Worst-regime CVaR_α is $\mathrm{WCVaR}_\alpha(\ell) = \max_{g=1,\ldots,G} \mathrm{CVaR}_\alpha(\{\ell_n : c_n = g\})$.

Regimes. Regime labels define groups for worst-regime CVaR_{95} in validation selection and test reporting; this metric tests whether tail-risk gains persist in stressed states rather than only in benign periods. They use market features available at episode start: Cboe Volatility Index (VIX), Cboe S&P 500 3-Month Volatility Index (VXV), at-the-money implied volatility (ATM IV), short IV, skew magnitude, term-slope magnitude, 5-day and 21-day backward realized volatility, downside semivolatility, and quote sparsity, measured as negative quote density. We construct $G = 3$ regimes by centering these features at training-split medians, scaling by training-split standard deviations, forming a weighted stress score, and applying train-split score tertile cutpoints to validation and test episodes. The stress score uses fixed weights that emphasize volatility, skew, realized-volatility, and quote-sparsity stress indicators; in the listed feature order, the weights are 1.0, 0.8, 1.0, 0.7, 0.8, 0.5, 0.8, 0.8, 0.7, and 0.35. The weights and tertile cutpoints are fixed before validation and test evaluation.

4 Method: Factor-Neutral Update Gating

When active hedge instruments outnumber first-order Greek constraints, delta-vega matching leaves degrees of freedom in hedge composition. FNUG uses these degrees of freedom by learning a bounded near-null update around the LSDeltaVega anchor. Such updates have small aggregate delta and scaled-vega exposures while changing the maturity, skew, and liquidity mix of hedge instruments. We decompose each hedge into the anchor position q_t^0 and a gated residual correction ρ_t; the final composition is defined in Eq. (6). The design operationalizes this idea by bounding the update, projecting it toward the delta-vega null space, gating execution, and penalizing anchor-relative underperformance. Figure 1 summarizes the resulting architecture.

4.1 Least-Squares Delta-Vega Anchor

Let $\Delta_{t,j}$ and $v_{t,j}$ be the delta and vega of hedge instrument j, and let Δ_t^L, v_t^L be the liability delta and vega. Define

$$A_t = \begin{bmatrix} \Delta_{t,1} & \cdots & \Delta_{t,M} \\ v_{t,1}/c_v & \cdots & v_{t,M}/c_v \end{bmatrix}, \qquad b_t = \begin{bmatrix} \Delta_t^L \\ v_t^L/c_v \end{bmatrix}, \tag{3}$$

where $c_v > 0$ is a fixed vega scale. Thus $A_t \in \mathbb{R}^{2\times M}$ and $b_t \in \mathbb{R}^2$. The deterministic base hedge used by all anchored policies is the clipped ridge LSDeltaVega anchor

$$\begin{aligned} \hat{q}_t^0 &= \arg\min_{q\in\mathbb{R}^M} \|A_t q - b_t\|_2^2 + q^\top R_t q, \\ q_t^0 &= \mathrm{clip}(\hat{q}_t^0, -q_{\max}, q_{\max}), \end{aligned} \tag{4}$$

where $q_{\max} > 0$ is the componentwise position limit and the clipping is componentwise. Here $R_t \in \mathbb{R}^{M\times M}$ is a positive definite regularization matrix containing ridge and spread-based penalties. Coordinates with $v_{t,j} = 0$ are excluded from the solve and set to $q_{t,j}^0 = 0$.

4.2 Policy Update and Factor-Neutral Projection

Let g_t collect the latent state z_t, liability descriptors, and volatility-index variables. The update network observes g_t together with candidate mark, spread, maturity, moneyness, and Greek features. The inputs also include the current LSDeltaVega anchor, the previous residual correction, and the availability mask. The network jointly outputs $(\tilde{u}_t, \eta_t, \xi_t) \in \mathbb{R}^{3M}$: an unconstrained residual target, activation-band logits, and proposal-weight logits. The target is bounded as $u_t^{\mathrm{tar}} = s \tanh(\tilde{u}_t)$, where s is the residual scale. Two mechanisms then temper this target before projection: a learned activation band suppresses small moves from the previous residual, and coordinatewise proposal weights control movement toward the banded proposal. Let ρ_{t-1} be the previous residual correction, with $\rho_{-1} = 0$. Define

$$\omega_t = s\ \mathrm{sigmoid}(\eta_t), \qquad m_t = \mathrm{sigmoid}(\xi_t),$$

where $\omega_t \in [0, s]^M$ is the learned activation band, $m_t \in [0, 1]^M$ is the coordinatewise proposal weight, and $\epsilon = 0.02$ is the activation temperature. All vector nonlinearities, absolute values, signs, and clipping operations below act coordinatewise; $\odot$ denotes the Hadamard product, $\mathrm{sigmoid}(x) = (1 + e^{-x})^{-1}$, and $\mathrm{softplus}(x) = \log(1 + e^x)$. With $e_t = u_t^{\mathrm{tar}} - \rho_{t-1}$ and $\mathrm{sign}(0) = 0$, the activated proposal is

$$\begin{aligned} u_t^{\mathrm{band}} &= \mathrm{clip}\left(\rho_{t-1} + \mathrm{sign}(e_t)\epsilon\ \mathrm{softplus}\left(\frac{|e_t| - \omega_t}{\epsilon}\right), -s, s\right), \\ \bar{u}_t &= \mathrm{clip}\Big(\rho_{t-1} + m_t \odot (u_t^{\mathrm{band}} - \rho_{t-1}), -s, s\Big). \end{aligned}$$

This map attenuates small residual gaps but does not impose an exact no-trade region; m_t interpolates coordinatewise toward the activated proposal, whereas the scalar trust gate below acts after projection. The resulting $\bar{u}_t$ is the pre-projection residual candidate. Before projection we set $\bar{u}_t \leftarrow v_t \odot \bar{u}_t$ and zero unavailable columns of A_t, then attenuate first-order delta-vega exposure by applying

$$r_t^{\mathrm{proj}} = \left[I_M - A_t^\top (A_t A_t^\top + \lambda_p I_2)^{-1} A_t\right] \bar{u}_t. \tag{5}$$

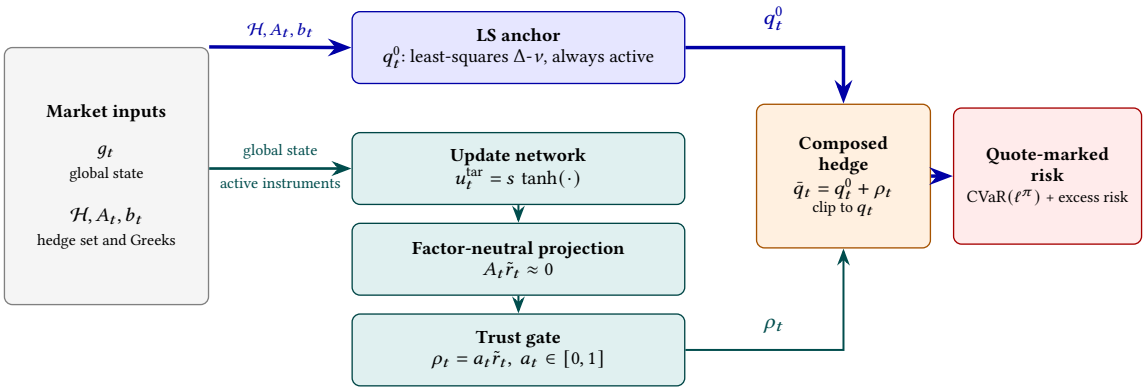


Figure 1: FNUG architecture. The LSDeltaVega branch (thick blue) uses the hedge set and delta-vega targets to compute the always-active anchor, while the neural branch (teal) uses the global state and active instruments to produce a bounded update through factor-neutral projection and trust gating. The two branches form $\bar{q}_t = q_t^0 + \rho_t$, which is clipped to the final hedge q_t. The resulting policy is trained using quote-marked P&L and excess risk relative to LSDeltaVega. When the gate is zero, $\rho_t = 0$; since q_t^0 already satisfies the position limit, the policy reverts to LSDeltaVega.

Here I_M and I_2 are identity matrices of dimensions M and 2, respectively, and $\lambda_p \geq 0$ is the projection regularization parameter. For $\lambda_p > 0$, Eq. (5) is a ridge-regularized null-space filter rather than an exact orthogonal projection, with

$$A_t r_t^{\text{proj}} = \lambda_p (A_t A_t^\top + \lambda_p I_2)^{-1} A_t \bar{u}_t.$$

When $\lambda_p = 0$ and $A_t A_t^\top$ is nonsingular, it reduces to the orthogonal projection onto $\ker(A_t)$. The remaining residual delta-vega exposure is measured during training and reported in Section 6.4.

After projection, the projected vector is scaled if necessary to define $\tilde{r}_t = c_t r_t^{\text{proj}}$, where $c_t = \min\{1, s/\|r_t^{\text{proj}}\|_\infty\}$ when $r_t^{\text{proj}} \neq 0$ and $c_t = 1$ otherwise. This keeps the largest absolute coordinate within the residual scale s without applying coordinatewise clipping after projection; finally $\tilde{r}_t \leftarrow v_t \odot \tilde{r}_t$.

We write the projected residual exposure used in training and diagnostics as $\chi_t^{\text{proj}} = A_t \tilde{r}_t = [\Delta_t^{\text{res}}, v_t^{\text{res}}/c_v]^\top$.

The update channel has room to act when active hedge coordinates outnumber Greek constraints. Let $n_t = \sum_j v_{t,j}$ and let A_t^{act} be the active Greek matrix; in the zero-ridge case, the dimension of the null space is $n_t - \text{rank}(A_t^{\text{act}})$. With two Greek constraints, the main $M = 4$ hedge set typically leaves a two-dimensional null space for the update to act in.

4.3 Trust Gate and Hedge Composition

The scalar trust gate is $a_t = \text{sigmoid}(g_\phi(g_t, \psi_t))$, where g_ϕ has parameters ϕ and ψ_t summarizes active instrument context. Each active instrument token contains candidate features, the anchor position $q_{t,j}^0$, and the previous residual $\rho_{t-1,j}$. Masked attention contextualizes these tokens, and ψ_t is their masked average. On active rebalance dates $t = 0, \ldots, H-1$, the residual correction, composed hedge before position limits, and final hedge are

$$\rho_t = a_t \tilde{r}_t, \qquad \bar{q}_t = q_t^0 + \rho_t, \qquad q_t = v_t \odot \text{clip}(\bar{q}_t, -q_{\max}, q_{\max}). \quad (6)$$

Terminal liquidation is imposed separately by setting $q_H = 0$. Because q_t^0 already satisfies the position limits, $a_t = 0$ or $\tilde{r}_t = 0$ gives $\rho_t = 0$ and recovers $q_t = q_t^0$ exactly.

4.4 Anchor-Relative Training Objective

The main risk terms control the policy's own tail loss and its tail underperformance relative to the anchor. For a learned policy π, let ℓ^π be its normalized episode loss and let ℓ^0 be the normalized episode loss of the LSDeltaVega anchor on the same episodes. The excess loss $\ell^\pi - \ell^0$ is the anchor-relative loss: it measures whether the learned update worsens the same historical episode relative to the classical hedge. Let $\mathbb{E}_{\text{ep}}$ average over training episodes and $\mathbb{E}_{\text{ep},t} = H^{-1}\mathbb{E}_{\text{ep}} \sum_{t=0}^{H-1}$. The core loss contains

$$\begin{aligned}\mathcal{L}_{\text{core}}(\pi) = \text{CVaR}_\alpha(\ell^\pi) + \beta\,\text{CVaR}_\alpha(\ell^\pi - \ell^0) + \gamma\,\mathbb{E}_{\text{ep}}\left[(\ell^\pi - \ell^0)_+\right] \\ + \lambda_\rho \mathbb{E}_{\text{ep},t}\|\rho_t\|_2^2 + \lambda_{\text{chg}}\mathbb{E}_{\text{ep}} \sum_{t=0}^{H-1} \|\rho_t - \rho_{t-1}\|_1 \\ + \lambda_A \mathbb{E}_{\text{ep},t}\|\chi_t^{\text{proj}}\|_2^2 + \lambda_g \mathbb{E}_{\text{ep},t} a_t + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}.\end{aligned} \quad (7)$$

Here π denotes the learned policy, α is the CVaR confidence level, and $(x)_+ = \max\{x, 0\}$. The weights $\beta, \gamma, \lambda_\rho, \lambda_{\text{chg}}, \lambda_A, \lambda_g$, and λ_{aux} are nonnegative hyperparameters, and $\mathcal{L}_{\text{aux}}$ is the auxiliary loss. The λ_ρ, λ_{chg}, and λ_g terms penalize residual size, residual turnover, and gate activity, respectively. Both CVaR terms use the Rockafellar–Uryasev minibatch surrogate [23], with a learned threshold for policy loss and a stop-gradient minibatch quantile for paired excess loss. The factor-exposure penalty is applied to χ_t^{proj}, before the scalar trust gate, so exposure control is enforced on the projected residual candidate during training. The actual training objective is $\mathcal{J}(\pi) = \mathcal{L}_{\text{core}}(\pi) + \mathcal{L}_{\text{oper}}(\pi)$, where $\mathcal{L}_{\text{oper}}$ contains fixed operating penalties for mean loss, transaction cost, turnover, position size, and anchor imitation, together with a sparsity-promoting log-sum penalty on hedge trades; Section 5 gives the main values.

$\mathcal{L}_{\text{aux}}$ is the mean Huber loss ($\delta = 0.25$) for one-step normalized anchor loss, normalized liability loss, absolute underlying return, and normalized anchor transaction cost. It provides dense supervision for the shared policy representation. Training begins with imitation of LSDeltaVega and then fine-tunes under the CVaR objective.

4.5 Training and Evaluation Protocol

The full procedure is summarized below.

(1) Build a contract-level option table with exact date, expiration, strike, option type key, midpoint, bid, ask, implied volatility, delta, vega, spread, and liquidity.
(2) Extract daily latent states and fixed implied volatility grid representations using only same-day option information.
(3) Construct episodes and hedge sets under the empirical design protocol. Instrument selection uses only episode-start information, and policy states use only same-day information.
(4) For every rebalance date, compute q_t^0 by Eq. (4); for FNUG, compute the projected residual candidate by Eq. (5), apply the trust gate, and form the hedge by Eq. (6).
(5) Select learned policies on validation data using the model-specific risk and feasibility rules in Section 5, then report test metrics using historical quote-marked P&L.

At each rebalance date, the policy state and transaction marks are taken from the same end-of-day snapshot. The backtest interprets the action as an immediate end-of-day rebalance and does not model intraday execution latency.

5 Empirical Design

Data. We use OptionMetrics IvyDB US end-of-day SPX option-chain data from 2018–2025 with bids, asks, implied volatilities, strikes, expirations, and option type flags. We retain European-style, PM-settled contracts with valid bid-ask quotes and positive implied volatility. Relative spread is $(\text{ask}-\text{bid})/\text{mid}$, where mid is the quote midpoint. Liability candidates require bid ≥ 0.10 and relative spread ≤ 0.30; all contract-path marks require bid ≥ 0.05 and relative spread ≤ 0.50. Delta and vega are recomputed from the midquote-implied volatility used in the hedging table. Path marks require log-moneyness $k \in [-0.60, 0.60]$, and quote-set preprocessing uses $k \in [-0.30, 0.30]$. Auxiliary data include the underlying index close, zero-coupon yield curve, dividend yield curve, VIX, and VXV. Splits are chronological: 2018–2021 for training, 2022 for validation, and 2023–2025 for testing.

Episodes. We evaluate three liability families: ATM call, 25-delta put, and ATM straddle. Liabilities target 30 days to maturity within a ±10-day window; ATM contracts minimize absolute log-moneyness, and put liabilities minimize distance to delta −0.25. Each episode has horizon $H = 5$ trading days and stride five in the main results. For each test family, this gives 133 test episodes. The main specification uses $M = 4$ instruments, cost levels 1, 2, 5 bps, and seeds 1, 42, 2026.

Hedge instrument selection. The underlying is always included. After excluding the liability contract or its straddle legs, we form 30-day candidates from ATM, 10-delta, and 25-delta calls and puts plus ATM straddles, and 60-day candidates from 25-delta calls and puts plus ATM straddles. Non-underlying candidates are ranked by start-date Black–Scholes vega divided by $(\text{relative spread} + 10^{-4})$, and the top three are retained; episodes require at least three active hedge coordinates.

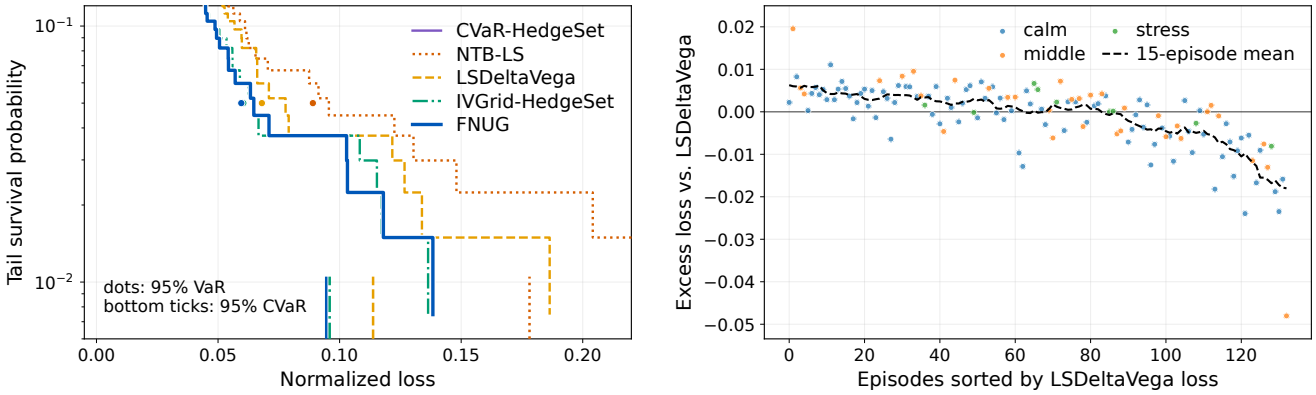
Baselines. All baselines use the same episodes, available hedge instruments, historical marks, and transaction-cost accounting as FNUG. Classical baselines include SmiledDelta, NoTradeDelta, GreedyDeltaVega, and LSDeltaVega. These are, respectively, a smile-implied delta hedge, a validation-selected no-trade delta hedge, a myopic hedge that reduces current delta-vega mismatch, and the least-squares delta-vega anchor in Eq. (4). The recent comparators are IVGrid-HedgeSet, which uses a fixed 16×21 grid over maturity and log-moneyness, with an observation-mask channel, and shares the LSDeltaVega-anchored factor-neutral update architecture; NTB-LS, which learns no-trade-band control around LSDeltaVega; CVaR-HedgeSet, an unanchored latent-state CVaR policy; and QiaoResidual-LS, a plain anchored residual learner adapted to our hedge-set protocol. QiaoResidual-LS keeps the same LSDeltaVega anchor but omits factor-neutral projection, trust gating, anchor-relative CVaR training, and auxiliary risk supervision. Together, these learned comparators separate state representation, the added control stack, trading-band control, and unanchored tail-risk learning.

Leakage controls. Episodes crossing data splits are discarded. Liability and hedge contracts are fixed by exact keys at episode start and retained only when quoted on all $H + 1$ dates, which rules out mid-episode substitution. Of the 383 windows eligible at episode start for each liability family, 382 are retained; the only exclusion is a training window that fails the complete-path requirement. This complete-path restriction may favor more persistent and liquid contracts. Policy normalizers and regime cutpoints use training data only; test episodes do not affect hyperparameters, model selection, or stopping.

Training protocol. All learned hedging policies are trained on the train split and selected on validation data. The main FNUG uses $\alpha = 0.95$, a frozen set encoder with latent dimension $d_z = 16$, hedge-policy width 96, three hidden layers, 10 imitation epochs, batch size 48, residual scale $s = 0.10$, $q_{\max} = 3.0$, $c_\nu = 100$, and projection regularization parameter $\lambda_p = 10^{-4}$. The LS regularizer R_t is diagonal with $[R_t]_{jj} = 10^{-3} + 0.05\, h_{t,j}/\bar{L}_0$. The main loss uses $\beta = 0.50$, $\gamma = 0.25$, $\lambda_\rho = 0.05$, $\lambda_{\text{chg}} = 0.02$, $\lambda_A = 0.10$, $\lambda_g = 0.003$, and $\lambda_{\text{aux}} = 0.05$. Auxiliary targets are one-step anchor loss, liability loss, absolute underlying return, and anchor TC. During the imitation warm start, the anchor-imitation weight is 1.0; the weights on turnover, position size, and the log-sum trade penalty are 0.05, 0.01, and 0.01, respectively. During CVaR fine tuning, the weights for episode mean normalized loss, normalized total TC, turnover, mean squared position, the log-sum trade penalty $\sum_j \log(1 + |q_{t,j} - q_{t-1,j}|/10^{-3})$, and anchor imitation $\|q_t - q_t^0\|_2^2$ are 0.15, 0.05, 0.02, 0.001, 0.002, and 0.02. The position and log-sum trade penalties are averaged over episode dates, while P&L and TC terms are normalized by $\bar{L}_0$. Optimization uses AdamW (learning rate 3×10^{-4}, weight decay 10^{-4}), cosine decay, unit gradient clipping, at most 250 epochs, and patience 35. The trust-gate bias is −1.0. Each liability-family, transaction-cost, and seed cell trains a separate learned policy; the global architecture and loss-hyperparameter configuration is held constant across these cells. For final selection, FNUG first requires validation excess $\text{CVaR}_{95} \leq 0.008$ and residual-exposure caps of 10^{-4} and then minimizes $\text{CVaR}_{95} + 0.25\text{CVaR}_{99} + 0.25\text{WCVaR}_{95}$. Learned comparators are selected by their validation scores and

Table 1: Main quote-marked test results. Losses and transaction costs (TC) are normalized by $\bar{L}_0 = \max\{L_0, 1\}$. Lower is better.

Family	Model	Mean	$CVaR_{95}$	$CVaR_{99}$	$WCVaR_{95}$	TC	Turnover
ATM call	FNUG	0.0020	**0.0945**	**0.1282**	**0.1015**	0.0263	2.105
	LSDeltaVega	0.0030	0.1138	0.1601	0.1321	0.0269	2.179
	GreedyDeltaVega	-0.0041	0.1633	0.2636	0.2109	0.0245	1.890
	SmiledDelta	-0.0104	0.3682	0.6365	0.5555	0.0176	1.318
	NoTradeDelta	-0.0097	0.3752	0.6402	0.5592	0.0175	1.305
25-delta put	FNUG	0.0421	**0.4138**	**0.8568**	**0.6971**	0.0362	1.412
	LSDeltaVega	0.0436	0.4330	0.8800	0.7193	0.0368	1.421
	GreedyDeltaVega	0.0454	0.5042	0.9998	0.8346	0.0343	1.217
	SmiledDelta	0.0326	0.8658	1.8153	1.5405	0.0191	0.660
	NoTradeDelta	0.0321	0.8926	1.7167	1.4652	0.0168	0.579
ATM straddle	FNUG	0.0141	**0.1177**	**0.2138**	**0.1577**	0.0144	2.311
	LSDeltaVega	0.0136	0.1319	0.2404	0.1798	0.0141	2.317
	GreedyDeltaVega	0.0106	0.1768	0.3211	0.2422	0.0118	1.903
	SmiledDelta	0.0048	0.3942	0.7034	0.5965	0.0054	0.825
	NoTradeDelta	0.0051	0.3941	0.7070	0.5947	0.0052	0.807



(a) Zoomed right-tail survival

(b) Paired excess loss against LSDeltaVega

Figure 2: Tail and anchor-relative evidence for ATM call liabilities at 2 bps and seed 42. Panel (a) shows the zoomed right-tail survival curve of normalized losses. Panel (b) plots paired excess loss, $\ell_i^{\text{FNUG}} - \ell_i^{\text{LSDeltaVega}}$, after sorting episodes by LSDeltaVega loss. Colors denote train-calibrated regimes; the dashed line is a 15-episode moving average over this ranking. Negative values favor FNUG.

otherwise use the same split, cost protocol, and optimization discipline.

Statistical comparisons. We use a centered, two-sided paired bootstrap over episodes with 2,000 resamples for broad model comparisons [7]. The ATM-call, 2-bps, seed-42 cell is the focal comparison: 2 bps is the middle cost level and seed 42 is the default seed; the remaining 26 cells are robustness checks. For this $CVaR_{95}$ comparison, we also use a paired moving-block bootstrap (MBB) with 10,000 resamples on date-ordered episodes [16] to account for serial dependence in episode losses across adjacent start dates, applying the same overlapping blocks to both models and recomputing CVaR in each draw. The focal block length is five episodes, about one trading month at stride five; lengths three and ten check whether the conclusion depends on block length. MBB intervals are percentile 95% confidence intervals and use the same two-sided centered p-value rule. Negative differences favor FNUG. Because each test family contains 133 episodes, empirical $CVaR_{99}$ depends on the worst one to two episodes. We therefore treat $CVaR_{99}$ as directional extreme-tail evidence and use $CVaR_{95}$, which averages roughly the worst seven episodes, as the primary tail metric. When reporting bootstrap differences, D denotes the first method minus the comparator.

6 Results

6.1 Primary Historical Hedging Results

Table 1 reports the core 2-bps, seed-42 results. Relative to LSDeltaVega, FNUG lowers $CVaR_{95}$ by 16.9% for ATM calls, 4.4% for 25-delta puts, and 10.8% for ATM straddles. The main effect is tail-risk reshaping rather than higher trading intensity: turnover is no higher than LSDeltaVega in any liability family, normalized TC differs from LSDeltaVega by at most 0.0006, and mean-loss differences are small and mixed.

Figure 2 shows the tail effect at two resolutions: Panel (a) visualizes it at the survival-curve scale, with the FNUG loss tail sitting below LSDeltaVega near and beyond the 95% quantile, while Panel (b) displays paired excess losses over the realized LSDeltaVega-loss ranking.

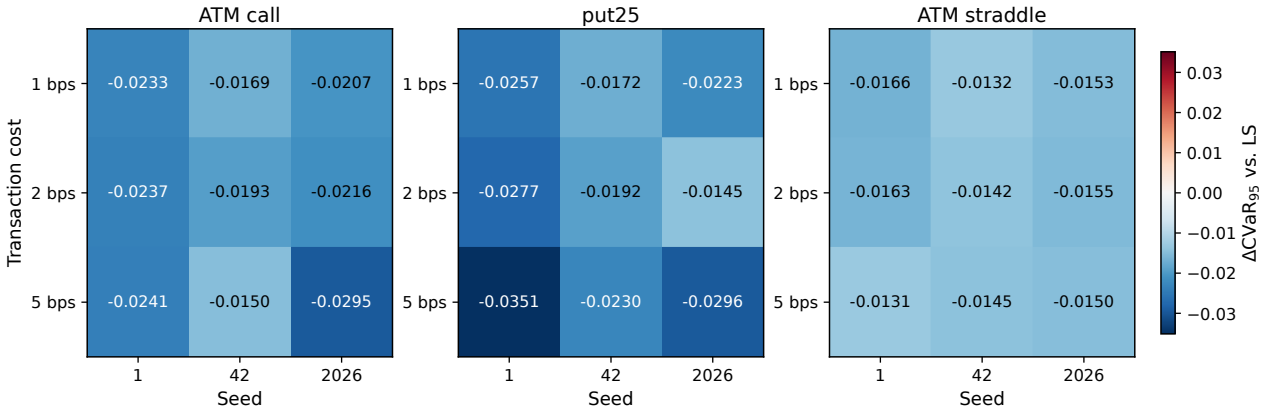


Figure 3: Cost and seed robustness across liability families. Each heatmap entry reports CVaR_{95}(FNUG) − CVaR_{95}(LSDeltaVega); negative values favor FNUG. Rows are transaction costs and columns are random seeds. For each family-cost pair, LSDelta Vega is deterministic on the fixed episode set, so the same LS value is used across seed columns.

For the ATM-call primary metric, the five-episode MBB gives $D = -0.0193$, a 95% confidence interval from -0.0309 to -0.0093, and $p = 0.0020$; the corresponding confidence intervals for block lengths three and ten also lie below zero. The CVaR_{99} point-estimate reduction is 0.0319; worst-regime CVaR_{95} also improves by 0.0306.

6.2 Fixed-Grid and Anchored-Residual Comparisons

Table 2 compares FNUG to modern deep or structural hedging baselines and to the closest anchored-residual comparator in the primary ATM-call setting with 2 bps transaction costs. Across the 27 matched IVGrid evaluations, FNUG has lower CVaR_{95} in 22 settings, with an average difference of -0.0064. The primary fixed-grid comparison is not decisive: the five-episode MBB gives $D = -0.0014$, a confidence interval of $[-0.0095, 0.0059]$, and $p = 0.6719$, with the same conclusion for block lengths three and ten. Relative to this strong fixed-grid competitor, FNUG delivers lower turnover (2.105 versus 2.139) and lower transaction cost (0.02634 versus 0.02659) in the displayed setting; paired bootstrap supports both differences ($p < 0.001$ and $p = 0.0005$, respectively). Thus, IVGrid gives comparable primary tail risk in the focal cell, while FNUG trades less and has a more favorable descriptive cross-setting CVaR_{95} pattern.

Against QiaoResidual-LS, the closest residual-learning comparator, the primary CVaR_{95} improvement is supported by the five-episode MBB with $D = -0.0247$, a 95% confidence interval from -0.0344 to -0.0066, and $p = 0.0018$; block lengths three and ten give the same conclusion. NTB-LS reduces trading but has worse tail losses, and the unanchored CVaR-HedgeSet performs poorly out of sample under the same quote-marked protocol.

Table 2: Fixed-grid, anchored-residual, and modern baselines for ATM call liabilities at 2 bps and seed 42. Exc_{95} is paired-underperformance CVaR; it is zero for LSDeltaVega by definition.

Model	CVaR_{95}	CVaR_{99}	WCVaR_{95}	Exc_{95}	Turnover
FNUG	**0.0945**	0.1282	0.1015	0.0102	2.105
IVGrid-HedgeSet	0.0959	**0.1268**	0.1015	0.0098	2.139
QiaoResidual-LS	0.1192	0.1412	0.1198	0.0274	2.192
LSDeltaVega	0.1138	0.1601	0.1321	0.0000	2.179
NTB-LS	0.1781	0.3295	0.3014	0.0877	1.958
CVaR-HedgeSet	0.6490	0.9555	0.9555	1.1341	0.806

6.3 Robustness Across Costs and Seeds

Figure 3 reports CVaR_{95}(FNUG) − CVaR_{95}(LSDeltaVega) across costs, seeds, and liability families. All 27 entries are negative, with average differences of -0.0216 for ATM calls, -0.0238 for puts, and -0.0149 for straddles, reducing the likelihood that the result is driven by a particular seed or cost assumption. A fixed local sensitivity audit perturbs anchor-risk and nonstructural soft-regularization weights around the final configuration to check that the gain does not hinge on exact loss-weight choices; the audit is separate from model selection, and all 12 perturbed runs remain below LSDelta Vega in CVaR_{95}. Across all 27 FNUG evaluation cells, the final position limits never bind; projected residual delta and scaled-vega exposures have maxima of 5.15×10^{-6} and 4.74×10^{-7}, respectively.

To assess statistical support for the average improvement, Table 3 reports $\bar{D}$, the mean of the CVaR_{95} differences D (FNUG minus the comparator) over the 27 combinations of liability family, cost, and seed. Each combination receives equal weight. Each MBB sample uses the same blocks of consecutive episodes for all settings; CVaR_{95} is recomputed for each model before averaging the differences.

Table 3: Across 27 settings, counts indicate lower FNUG CVaR_{95} point estimates. $\bar{D}$ is the mean CVaR_{95} difference (FNUG minus the comparator) over these settings. MBB uses blocks of five episodes and 95% percentile confidence intervals (CIs). p_{H} denotes the p value after Holm adjustment for the three baseline comparisons.

Comparator	Lower CVaR_{95}	$\bar{D}$	95% CI	p_{H}
LSDeltaVega	27/27	−0.0201	[−0.0340, −0.0087]	**0.0342**
IVGrid	22/27	−0.0064	[−0.0157, 0.0005]	0.1904
Qiao-LS	26/27	−0.0210	[−0.0443, 0.0122]	0.1904

IVGrid: IVGrid-HedgeSet; Qiao-LS: QiaoResidual-LS.

With blocks of five episodes, the average reduction in CVaR_{95} relative to LSDeltaVega is significant after adjustment using Holm's method for the three baseline comparisons [13]. The confidence intervals for the mean differences relative to IVGrid-HedgeSet and QiaoResidual-LS include zero. Blocks of three and ten episodes give the same conclusions.

6.4 Ablation and Exposure Evidence

Table 4 reports a three-seed component audit for the ATM-call, 2-bps setting. The no-factor-neutral variant removes the projection and factor-exposure penalty. For this variant, we select checkpoints by validation score without feasibility filtering. The no-trust-gate and no-auxiliary-risk variants fix $a_t = 1$ and set $\lambda_{\mathrm{aux}} = 0$, respectively. All variants otherwise use the same operating-penalty settings.

The no-factor-neutral variant shows the largest degradation in this audit: it raises CVaR_{95} from 0.0922 to 0.1106 and increases residual Greek exposures by several orders of magnitude. Removing the scalar trust gate a_t or auxiliary supervision changes tail risk only modestly, suggesting that these components are secondary in this ATM-call audit.

Table 4: ATM-call component evidence at 2 bps, averaged over seeds 1, 42, 2026. Exc_{95} is paired-underperformance CVaR; residual Greek columns report mean absolute projected exposures from $A_t\tilde{r}_t$, and Trust is the mean gate value.

Model	CVaR_{95}	CVaR_{99}	WCVaR_{95}	Exc_{95}	$\lvert\Delta^{\mathrm{res}}\rvert$	$\lvert\nu^{\mathrm{res}}\rvert/c_\nu$	Trust
FNUG	0.0922	0.1258	0.1005	0.0118	2.4×10^{-6}	2.0×10^{-7}	0.695
Qiao-LS	0.1146	0.1505	0.1422	0.0387	1.1×10^{-2}	2.3×10^{-1}	–
No FN	0.1106	0.1520	0.1317	0.0322	3.9×10^{-2}	4.9×10^{-1}	0.318
No gate	0.0939	0.1262	0.1000	0.0115	3.6×10^{-6}	2.5×10^{-7}	1.000
No Aux	0.0932	0.1260	0.1003	0.0110	2.6×10^{-6}	2.3×10^{-7}	0.692
LS only	0.1138	0.1601	0.1321	0.0000	0	0	–

Qiao-LS: QiaoResidual-LS; FN: factor-neutral control; Aux: auxiliary risk; LS: LSDeltaVega.

To check whether the component comparisons depend on model selection, we reselect FNUG and the three ablation variants from the training runs used in Table 4. All four models use the composite validation score originally used for the no-factor-neutral variant, with identical weights and no validation thresholds on excess CVaR_{95} or residual Greek exposure. The score definition and fixed weights are available in the code repository listed under Code availability. We then compare the selected models on the test episodes using paired MBB, applying the same blocks of consecutive episodes to both compared models and all three seeds in each sample. We use blocks of five episodes for the main analysis, with lengths three and ten as sensitivity checks.

Table 5 reports mean CVaR_{95} and bootstrap comparisons of FNUG with the three ablation variants under the common selection rule. FNUG has 18.7% lower mean CVaR_{95} than the no-factor-neutral variant. The difference is significant after adjustment using Holm's method for the three component comparisons. The conclusions are unchanged across the tested block lengths. The no-trust-gate variant has a lower CVaR_{95} point estimate than FNUG, while the confidence intervals for the no-trust-gate and no-auxiliary-risk comparisons both include zero.

Table 5: ATM-call component comparison at 2 bps, with selection by the recorded validation score and no feasibility thresholds. CVaR_{95} is averaged over seeds 1, 42, 2026; $\bar{D}$ is the mean CVaR_{95} difference over these seeds (FNUG minus each variant). MBB uses blocks of five episodes and 95% percentile CIs. p_{H} denotes the p value after Holm adjustment for the three component comparisons.

Model	CVaR_{95}	$\bar{D}$	95% CI	p_{H}
FNUG	0.0899	–	–	–
No FN	0.1106	−0.0207	[−0.0325, −0.0100]	**0.0009**
No gate	0.0867	0.0033	[−0.0023, 0.0093]	0.2544
No Aux	0.0920	−0.0021	[−0.0047, 0.0003]	0.1636

7 Conclusion

We introduced FNUG as a controlled policy-improvement framework for option hedging with irregular quote sets. When active instruments outnumber first-order constraints, LSDeltaVega resolves the resulting indeterminacy in hedge composition through ridge and spread penalties. FNUG instead turns those degrees of freedom into a learned decision space, using anchor-relative tail risk to select bounded, gated refinements of the classical hedge. Historical SPX results show lower tail risk than LSDeltaVega and a plain residual learner without higher trading intensity. Within this framework, the quote-set and fixed-IV-grid policies deliver comparable primary tail risk in the focal comparison. The strongest evidence points to the factor-neutral update design; the quote-set policy is descriptively favored, while an encoder-specific advantage remains unestablished.

The no-factor-neutral variant shows the largest deterioration among the tested variants. Projected residual delta and scaled-vega exposures are of order 10^{-6} and 10^{-7}, respectively, indicating that the gain is achieved by reallocating positions across hedge instruments while preserving first-order control, not by relaxing it. The trust gate and auxiliary task have smaller incremental effects in this audit, while the respective roles of higher-order, dynamic, and liquidity channels in the gain remain to be disentangled.

This interpretation yields a testable hypothesis: gains should increase with the active null-space dimension and diminish when gamma is added as a third Greek constraint. Future work can test this hypothesis and extend the framework to single-name options, less liquid markets, intraday hedging, 0DTE contracts, and live execution with a tradable underlying proxy.

Code availability. Code and derived analysis outputs are available at https://github.com/TenghanZhong/Fnug_code.

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